Agentic Engineer Capstone -- Rubric Self-Check

Purpose

This self-check maps the current capstone evidence to the Canvas scoring criteria.

Statuses are intentionally conservative:

* **Complete** means repository evidence currently supports the criterion.

* **Partial** means meaningful evidence exists but additional final evidence is still needed.

* **Blocked** means completion currently depends on an external runtime dependency.

* **Pending** means the artifact or evidence has not yet been completed.

No criterion is marked complete based on estimated or invented results.

1. Workflow Scoping

Status: Complete

Evidence:

* docs/capstone-scope.md

* Stakeholder identified

* Trigger defined

* Inputs and outputs defined

* Acceptance criteria defined

* Failure modes identified

* Job-seeker/no-deployment path selected

* Existing Module 1-4 evidence and gaps inventoried

The selected workflow is an AI-assisted code review and governance pipeline with measurable review, quality, governance, and cost goals.

2. Sandboxed Environment

Status: Complete

Evidence:

* Dockerfile

* docker-entrypoint.sh

* scripts/run-agent.sh

* Role-specific read-only/read-write container behavior

* Memory mounted only where required

* Enforcement evidence in eval/enforcement-verification.md

The system runs in a containerized environment with explicit role-based workspace and memory boundaries.

The README now documents the capstone run path, required model credential, governed agents, preserved final-run evidence, and operational limitations for a new reviewer.

3. Quality Spec & Baseline

Status: Partial

Evidence:

- docs/prd.md
- eval/rubric.json
- historical Module 1 PRD/rubric/iteration evidence
- docs/calibration-log.md
- docs/capstone-impact-report.md
- docs/run-summary.md
- logs/capstone-final-human-approved-run.json

The rubric defines four judgment dimensions with pass thresholds.

Measured baseline and comparison evidence exists for the deterministic conversion, including cycle time, model-token cost, review latency, structural checks, and repeatability.

The successful final production-like run also provides final execution evidence, including a 59.6-second full-pipeline runtime and recorded Planner, Reviewer, and Tester evaluation outcomes.

Limitation

The available historical Module 1 artifacts do not provide one complete baseline containing every capstone impact metric.

In addition, the final transcript does not establish a trustworthy full-pipeline model-cost measurement, and one successful run is not sufficient evidence for a general defect-rate percentage.

The impact report therefore keeps historical baseline evidence, deterministic-conversion measurements, and final production-like measurements separate rather than reconstructing or inventing missing values.

4. Agents, Skills & Memory

Status: Complete

Evidence:

* versioned role definitions under agents/

* persistent state in .memory/storage.db

* storage audit evidence

* reusable references under .memory/reference/

* role-scoped memory mounting

* docs/adr/ADR-004-memory-layout.md

The memory architecture separates persistent state, reusable references, task context, and role instructions.

5. Orchestration & MCP Tools

Status: Complete

Evidence:

* docs/orchestration-diagram.md

* docs/routing-and-tool-grant-map.md

* mcp-servers/storage/

* mcp-servers/retrieval/

* role-specific allow-lists

* classification ceilings

* retrieval ground-truth set

* retrieval quality report

* schemas/handoff.json

The Orchestrator delegates to scoped roles and MCP capabilities are not exposed globally.

6. Evaluation & Calibration

Status: Complete

Evidence:

- eval/test_deterministic.py
- eval/test_rubric_suite.py
- eval/rubric.json
- docs/holdout-task-set.md
- docs/calibration-log.md
- eval/run_holdout.py
- eval/run_regression.py
- regression and policy tests
- retrieval validation evidence
- logs/capstone-final-human-approved-run.json
- docs/run-summary.md

The project uses a two-layer evaluation design combining deterministic checks with judgment-based evaluation criteria.

Calibration and regression evidence includes:

- 7/7 deterministic checks passing before and after the handoff-validation conversion;
- zero output variance across three deterministic runs after conversion;
- 24/24 integrated regression checks passing;
- passing policy evidence;
- retrieval calibration and validation evidence;

• preserved holdout tasks;

• a successful production-like model-backed workflow run.

The final production-like run completed the Planner -> Implementer -> Reviewer -> Tester path and recorded successful evaluation outcomes.

Measurement limitations, including the lack of a trustworthy full-pipeline model-cost value and the inability to generalize a defect-rate percentage from one successful run, are documented separately in docs/capstone-impact-report.md.

7. Governance, Security & CI/CD

Status: Complete

Evidence:

* docs/governance-policy.md

* docs/routing-and-tool-grant-map.md

* MCP allow-lists

* data-classification ceilings

* eval/test_policy.py

* eval/test_governed_files.py

* eval/red-team-prompts.md

* eval/red-team-results.md

* eval/enforcement-verification.md

* .github/workflows/ci.yml

* docs/ci-step-design.md

The tool-evolution drill also demonstrated that permission drift is automatically detected by the policy suite.

Known platform-level limitation:

Direct pushes from the authorized repository owner can bypass the configured pull-request and required-check rules. This is documented separately from repository-level enforcement.

8. Right-Tool Decisions & ADRs

Status: Complete

Evidence:

* docs/step-classification.md

* deterministic handoff validation

* deterministic routing decision history

* before/after latency and cost measurements

* ADR package under docs/adr/

ADR coverage includes:

* rubric design;

* memory layout;

* MCP boundaries;

* subagent scoping and routing;

* governance policy;

* agent-to-deterministic conversion.

Measured handoff conversion evidence shows:

* 45 seconds to 0.2 seconds cycle time;

* $0.003 to $0 model-token cost;

* approximately 30 seconds to 5 seconds review latency;

* 7/7 structural checks preserved.

9. Production Integration & Tool-Evolution Drill

Status: Complete

Completed evidence:

- successful production-like model-backed orchestration;
- Planner -> Implementer -> Reviewer -> Tester execution;
- deterministic lexical retrieval with classification-ceiling enforcement and persistent-storage activity;
- Implementer storage write with downstream verification;
- governance enforcement;
- tool-evolution drill;
- regression detection and recovery;
- preserved transcript and audit evidence.

The successful final human-approved production-like run completed in 46.8 seconds using 33,519 of the bounded 38,000-token workflow budget. Human approval was approve and completion_authorized was true.

Evidence:

- logs/capstone-final-human-approved-run.json
- docs/run-summary.md
- docs/capstone-tool-evolution-drill.md

Tool-evolution result:

Stage	Result
Baseline	6 passed
Permission revoked	1 failed, 5 passed
Permission restored	6 passed

Governance enforcement was also demonstrated independently. An unauthorized documentation-writer write_entry request was denied by the storage MCP allow-list and recorded as authorization_denied.

The earlier capstone-final-run-001 is retained as evidence of an integration failure that was diagnosed and corrected before the successful final run.

10. Iteration Narrative & Impact

Status: Complete with documented measurement limitations

Evidence:

- docs/iteration-log.md
- docs/calibration-log.md
- docs/capstone-impact-report.md
- docs/run-summary.md
- deterministic conversion before/after evidence
- successful production-like run
- tool-evolution drill

The evidence shows multiple evidence-driven improvement cycles.

Measured deterministic-conversion impact includes:

- cycle time: 45 seconds -> 0.2 seconds;
- model-token cost: $0.003/run -> $0/run for the converted step;
- review latency: approximately 30 seconds -> approximately 5 seconds;
- structural quality: 7/7 checks passing before and after conversion;
- output variance: zero across three deterministic runs;
- integrated regression: 24/24 checks passing.

The final production-like pipeline completed in 59.6 seconds. In that successful run, 3 Planner review items, 4 Reviewer review items, and 1 Tester review item were approved, with no recorded rejected item.

Two measurement limitations remain explicit:

- one successful run is not sufficient evidence for a general 0% defect-rate claim;
- the transcript does not provide a trustworthy full-pipeline model-cost measurement.

No unsupported values are estimated or invented.

11. Stakeholder Communication

Status: Partial

Evidence:

- docs/capstone-stakeholder-one-pager.md
- docs/capstone-architecture-writeup.md
- docs/capstone-impact-report.md
- business-value framing
- risk-reduction explanation
- successful production-like run evidence in docs/run-summary.md

The stakeholder one-pager and supporting reports now describe the completed production-like workflow, measured deterministic-conversion impact, governance enforcement, business value, and remaining measurement limitations.

Remaining requirement:

- record the final five-to-ten-minute technical walkthrough demonstrating the completed workflow, governance enforcement, evaluation evidence, right-tool decision, measured impact, and operational readiness.

12. Clarity & Flow

Status: Partial

The planned final narrative follows:

1. Problem and baseline

2. Architecture

3. Live workflow

4. Governance

5. Evaluation

6. Right-tool decisions

7. Impact

8. Operations

Evidence documents already follow this narrative structure.

Final scoring depends on the completed walkthrough and presentation package.

13. Design

Status: Pending Final Packaging

Current content artifacts are complete enough to support final presentation design.

Remaining work includes:

* convert required documents to polished PDFs;

* ensure architecture diagrams and tables remain legible;

* create the final slide/video presentation;

* perform consistent visual formatting and sanitization.

Current Readiness Summary

Criterion	Status
Workflow Scoping	Complete
Sandboxed Environment	Complete
Quality Spec & Baseline	Partial — historical baseline limitations documented
Agents, Skills & Memory	Complete

Criterion	Status
Orchestration & MCP Tools	Complete
Evaluation & Calibration	Complete
Governance, Security & CI/CD	Complete
Right-Tool Decisions & ADRs	Complete
Production Integration & Tool-Evolution Drill	Complete
Iteration Narrative & Impact	Complete with documented measurement limitations
Stakeholder Communication	Partial — walkthrough video remaining
Clarity & Flow	Partial — final walkthrough/package remaining
Design	Pending Final Packaging

Remaining Critical Path

The remaining capstone critical path is now focused on final packaging and presentation rather than core workflow implementation:

1. perform the final sanitization and secret review;
2. run the final test and repository-status checks;
3. convert the required final documents to polished PDFs;
4. verify architecture diagrams and tables remain legible in the packaged artifacts;
5. record the five-to-ten-minute technical walkthrough;
6. perform the final submission review.

The production-like workflow, transcript and audit evidence, impact-report update, README run instructions, governance demonstration, and tool-evolution drill are already complete.

Self-Check Conclusion

The repository now contains substantial completed evidence for workflow design, sandboxed execution, specialized agents, memory architecture, orchestration, MCP boundaries, evaluation and calibration, governance, CI/CD controls, deterministic conversion, ADRs, production-like integration, and tool-evolution testing.

A successful production-like run is preserved in logs/capstone-final-human-approved-run.json with supporting audit evidence and a final run summary.

The remaining work is concentrated on final submission packaging and presentation rather than core system implementation.

Known limitations remain explicitly documented: the historical baseline does not contain every desired capstone metric, the final transcript does not establish a trustworthy full-pipeline model-cost measurement, one successful run is not sufficient to claim a general 0% defect rate, and the successful final run did not trigger the human-escalation path.

These limitations are reported rather than filled with estimated or invented results.